Properties of 2D Shapes - Mark scheme

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1

Edexcel IGCSE Mathematics A (4MA1) · Paper 2FR · June 2026 · Question 14

Question	Working	Answer	Mark	Notes
*	$360 \div 24$ oe or $\frac{(n-2)\times 180}{n} = 180 - 24$ oe		2	M1
	Correct answer scores full marks (unless from obvious incorrect working)	15		A1
				Total 2 marks

Question 2

Edexcel IGCSE Mathematics A (4MA1) · Paper 2F · June 2022 · Question 10

Question 14 June 2022 · Paper 2F · Q10 — **3 marks**

Part	Working / answer	Mark	Notes
	Area of kite = ½ × 6 × 4 (or counting squares)	**M1**	
	= 12 cm²	**A1**	
	Rectangle of area 12 cm² drawn, e.g. 4 × 3, 6 × 2	**B1**	ft their area

Question 3

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2017 · Question 8

8	$180 - 156$ (=24) **or** $180(n-2) = 156n$ oe **or** $90(2n-4) = 156n$ oe			M1
	$360 \div$ "24" **or** $(180 \times 2) \div (180 - 156)$ **or** $\frac{90\times 4}{2\times 90 - 156}$			M1 complete method
		15	3	A1
				Total 3 marks

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Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 2F · June 2026 · Question 23

Q	Working	Answer	Mark	Notes
	$180 \times (5-2)$ or $5 \times \left(180 - \frac{360}{5}\right)$ $(= 540)$ oe **OR** $(180-96)+(180-91)+(180-93)+2y = 360$ or $84+89+87+2y = 360$ **or** $(180-96)+(180-91)+(180-93)$ $(= 84+89+87 = 260)$ or $360-(180-96)-(180-91)-(180-93)\,(= 100)$		3	M1 for a correct method to find the sum of the interior angles **OR** forming an equation involving exterior angles (y is the exterior angle at E and B) may use any letter **or** finding the total of the exterior angles for the known angles or for the unknown angles
	eg $(2x =)$ "540" $-96-91-93$ $(=$ "540" $-280 = 260)$ or $(x =) \frac{"540"-96-91-93}{2}$ oe $\left(= \frac{"540"-280}{2}\right)$ **OR** $(y =) \frac{"100"}{2}$ $(=50)$ or $(x =) 180 - \frac{"100"}{2}$ oe			M1 (dep on M1) for getting as far as the value for $2x$ **or** for a complete method for x using interior angles **OR** for finding the exterior angle at E or B **or** for a complete method using exterior angles
	Correct answer scores full marks (unless from obvious incorrect working).	130		A1 cao
				Total 3 marks

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Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2023 · Question 11

11	eg $DEK = \frac{360}{9}$ **or** 40 **or** interior angle = $\frac{(9-2)\times 180}{9}$ **or** 140 or $OFK = 140 \div 2$ (= 70) or $FOK = \frac{2}{9}\times 360$ (= 80) or $EDK = 180 - 0.5 \times 140$ (=110) Angles marked correctly (any exterior or interior angle) gains this mark		3	M1	method to find interior or exterior angle or correct interior or exterior angle stated or shown correctly on diagram or for using 70° for *OFK* or 80° for *FOK* or 110 for *EDK* If a student has only found an interior or exterior angle and has clearly mixed up interior and exterior angles this mark cannot be awarded but can still award for any of the others angles stated
	$EDK = 110$ and $DEK = 40$ or $FOK = 80$ and $OFK = 70$ or $ODE = 70$ and $DEK = 40$ or $FED = 140$ and $EDK = 110$ oe			M1	For two correct angles that can lead directly to the answer in a single step (eg 180 – both angles or one angle minus the other)
	Correct answer scores full marks (unless from obvious incorrect working)	30		A1	
					Total 3 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2014 · Question 19

19.	180 ÷ (1 + 7) (= 22.5) 360 ÷ "22.5"	16	3	M1 M1ft dep M2 for $\frac{180(n-2)}{n} = \frac{7\times 180}{8}$ or $\frac{7\times 360}{n}$ or M2 for $360 \div \frac{180}{(1+7)}$ A1 cao
				Total 3 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 7

Question		Working	Answer	Mark	Notes
*	(a)	$360-85-60-90$ **or** $85+60+90 = 235$ **and** $360 - "235"$		2	M1
		Correct answer scores full marks (unless from obvious incorrect working)	125		A1
	(b)	$7a+5a=180$ **or** $12a=180$ **or** $(a=)\frac{180}{12}$ **or** $(a=)\frac{180}{7+5}$ **or** $(7a=)105$ **or** $(5a=)75$		2	M1
		Correct answer scores full mark (unless from obvious incorrect working)	15		A1
					Total 4 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2024 · Question 7

Question 10 November 2024 · Paper 1F · Q7 — **4 marks**

Part	Working / answer	Mark	Notes
	Angle ADC = 180 − 138 = 42°	M1	
	x = 360 − 125 − 107 − 42	M1	
	x = 86	A1	
	Reasons: angles on a straight line add up to 180°; angles in a quadrilateral add up to 360°	B1	Both reasons needed

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 2FR · January 2022 · Question 9

Question 16 January 2022 · Paper 2FR · Q9 — **4 marks**

Part	Working / answer	Mark	Notes
(i)	360 − 92 − 44 = 224	M1	
	224 ÷ 2	M1	
	x = 112	A1	
(ii)	Angles in a quadrilateral add up to 360°	B1	

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 15

Question	Working	Answer	Mark	Notes
	360 – 314 (= 46)		4	M1 angles may be marked on the diagram; M2 for $\frac{314-180}{2}$
	180 – "46" (= 134)			M1 do not award this mark for 314 – 180 (=134)
	Correct angle scores 3 marks (unless from obvious incorrect working)	67		A1
				B1 dep on M2 for two correct reasons from (i) angles around a point add up to 360° (ii) angles in a triangle add to 180° **or** angles in a triangle add to 180° (iii) base angle in an isosceles triangle (are equal) **OR** dep on M2 for The exterior angle of a triangle is equal to the sum of the interior opposite angles
				Total 4 marks

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Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2019 · Question 6

Question	Working	Answer	Mark	Notes
6	$360 \div 8\ (=45)$ or $180-(360 \div 8)\ (=135)$ or $\frac{6\times180}{8}\ (=135)$ oe e.g. $\frac{540-112-112-84}{2}\ (=116)$ or $\frac{540-308}{2}\ (=116)$ or $\frac{232}{2}\ (=116)$ e.g. '135' – '116' or 180 – '116' – '45'	19	4	M1 Correct method to find the interior or exterior angle of octagon M1 Correct method to find a missing angle from pentagon M1 Complete method A1

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2018 · Question 9

9	Interior angle of pentagon $(180 \times 3) \div 5\ (=108)$ oe		4	M1 or exterior angle of pentagon $=\frac{360}{5}\ (=72)$
	Interior angle of octagon $(180 \times 6) \div 8\ (=135)$ oe			M1 or exterior angle of octagon $=\frac{360}{8}\ (=45)$
	(*CBF* =) 360 – ("108" + "135") (= 117)			M1 (*CBF* =) "72" + "45" (= 117)
		31.5		A1
				Total 4 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2015 · Question 9

9. (a)	$(2 \times 7 - 4) \times 90$ or $(7-2) \times 180$ or $7\left(180-\frac{360}{7}\right)$ or $720+180$		2	M1	For the sum of the interior angles or an interior angle Eg Allow M1 for $\frac{(7-2)\times180}{7}$ oe or 128.(571428.....) correctly rounded or truncated to at least 3SF.
		900		A1	Mark the final answer
(b)	$\frac{1}{2}\left(180-\frac{900}{7}\right)$ or $0.5 \times (360/7)$ or $\frac{128.(571428\ldots)}{5}$		2	M1ft	For a complete and correct method
		25.7		A1	Accept 25.5 – 25.72 Allow $\frac{180}{7}$ or $25\frac{5}{7}$ oe
					Total 4 marks

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Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2022 · Question 11

11	$\frac{360}{10}$ (= 36) ext angle **or** $\frac{(10-2)\times 180}{10}$ (= 144)		4	M1	method to find interior or exterior angle. (angles may be seen on diagram)
	x = "144" – 90 (= 54) or $x = \frac{"540" - 3\times "144"}{2}$ (=54) or x = 90 – "36"(=54) 54 on the diagram is insufficient – must see working			M1	method to find x (must show it is intended to be x) eg use of int angle – 90º use of ext angle + x = 90º use of pentagon *GHIJA* All figures in " " must come from correct working
	$BAD = CDA = GDE = DGF = \frac{360 - 2\times "144"}{2}$ (= 36)			M1	A correct method to find an angle of 36º within the shape (not exterior angle) or 36º shown in correct place in diagram
	There are other correct methods. Please check for correct working.	x = 54 y = 54		A1	dep on M3 to find each of x and y and the correct value of 54 for both from correct working
					Total 4 marks
ALT	ADG = "144" – 2 × "36" (= 72)			M1	
	JA is parallel to GD			M1	
	$DGA = DAG$ (y) [isosceles triangle]			M1	
	$x = DGA = y$	shown		A1	
	There are other correct methods. Please check for correct working.				**Total 4 marks**

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2014 · Question 12

12.	(DBC =) 60 – x (Angles in an) equilateral triangle (= 60 degrees) BDC = 60 – x or BCD = 60 + 2x oe Base/bottom angles in an isosceles triangle (are equal) (BCD =) 60 + 2x	2x	4	B1 B1 B1 Can be marked on diagram. B1	Can be marked on diagram. {Reason 1} Can be marked on diagram. {Reason 2} both reasons 1 and 2 needed for B1 Answer only = B3. Numerical methods leading to a numerical answer can only score B1 (for giving both reasons adequately).
	Alternative: {Call ACD "y"} (BDC and DBC =) 60 – "y"/2 Base/bottom angles in an isosceles triangle (are equal) x + (60 – "y"/2) = 60 oe (Angles in an) equilateral triangle (= 60 degrees)	2x	4	B2 B1 B1	B2 for both (BDC and DBC =) 60 – y/2 B1 for either (BDC or DBC =) 60 – y/2 Can be marked on diagram. {Reason 1} i.e. Angle ABC is 60 {Reason 2} both reasons needed for B1 Answer only = B3. Numerical methods leading to a numerical answer can only score B1 (for giving both reasons adequately).
					Total 4 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2013 · Question 13

13	20 or 160 360 ÷ "20" or 720 ÷ "40"	18	4	M1 for (140 + x) + x = 180 oe or y + (y – 140) = 180 or $\frac{180(n-2)}{n} - 140 = \frac{360}{n}$ A1 x = 20 or y = 160 or 180n – 140n = 720 M1dep on previous M1 A1
				Total 4 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2016 · Question 13

13	$180-\frac{360}{10}$ or $\frac{(10-2)\times180}{10}$ or 144 oe		4	M1	Unless inconsistently labelled
	$\frac{180-'144'}{2}$ or 18			M1	Or M2 for 144 – (180 – 144)
	'144' – 2 × '18'			M1	
		108		A1	dep on M1
	Alternative				
	Pentagon approach – drawing in a pentagon or a statement recognising that the required angle is one of a regular pentagon		4	M1	May be implied by further work
	$180-\frac{360}{5}$ or $\frac{(5-2)\times180}{5}$			M2	(M1 for exterior angle of pentagon as long as not seen as interior angle or given as answer)
		108		A1	dep on M1
					Total 4 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 15

15	eg (6 – 2) × 180 (= 720)		4	M1	for a method to find the sum of the interior angles for a hexagon
	eg "720" – (90 + 95 + 149 + 104 + 57) (= 225)			M1	for a method to find the missing angle in the hexagon
	eg $\frac{360}{"225"-180}$ **or** $\frac{360}{"45"}$ **or** $\frac{180(n-2)}{n}=360-"225"$ oe **or** $\frac{180(n-2)}{n}="135"$			M1	for a complete method
	Working required	8		A1	cao dep on M2 NB: the answer of 8 can be gained from assuming that *AB* splits reflex *GBC* into 2 equal angles – without gaining the first 2 method marks [M0M0 is awarded] Award SCB1 for the student who gains an answer of 8 from this assumption or trial and improvement or no method shown
					Total 4 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2016 · Question 4

4	(a)		63	1	B1	
	(b)		50	1	B1	
	(c)	Eg (6 – 2) × 180 or 4 × 180 or 720 oe			M1	For complete method to find the total of interior angles or 720
		Eg $3x + x + 164 + 139 + 97 + 156 = 720$ or $4x + 556 = 720$ oe or $\frac{"720"-(164+139+97+156)}{4}$ or $\frac{"720"-556}{4}$ or $\frac{164}{4}$ oe			M1	Dep For a correct equation using their 720 or For a complete numerical method
			41	3	A1	
		Alternative Method Eg 180 – 156 + 180 – 139 + 180 – 164 + 180 – 97 + 180 – x + 180 – $3x$ = 360 or 24 + 41 + 16 + 83 + 180 – x + 180 – $3x$ = 360 or 1080 – 556 – $4x$ = 360			M2	For an equation coming from the correct method relating to the sum of exterior angles.
			41	3	A1	
						Total 5 marks

CHALLENGE QUESTIONS

Question 20 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2012 · Question 13*

13.	finds int angle of pentagon $\frac{(5-2)\times180}{5}$	finds ext angle of pentagon $\frac{360}{5}$		5	M1	for $\frac{(5-2)\times180}{5}$ or $\frac{360}{5}$	Award M1A1 for int angle of pentagon shown as 108° or ext angle shown as 72° on printed diagram or on candidate's own diagram
	108	72			A1	for 108 or 72	
	If there is *clear* evidence the candidate thinks the *interior* angle is 72° or the *exterior* angle is 108°, do not award the above two marks.						
	int angle of polygon = 144 or ext angle of polygon = 36				B1	for int angle of polygon = 144 or ext angle of polygon = 36	Award B1 for int angle of polygon shown as 144° or ext angle shown as 36° on printed diagram or candidate's own diagram
	$\frac{360}{36}$ or $\frac{180(n-2)}{n}=144$ oe				M1	for $\frac{360}{36}$ or $\frac{180(n-2)}{n}=144$ oe	
			10		A1	for 10 cao Award no marks for an answer of 10 with no working Award 5 marks for an answer of 10 if at least the first M1A1 are awarded	
							Total 5 marks

Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 12*

12	$9x+27+2x-23=180$ oe eg $11x+4=180$ **or** $\frac{180-27+23}{9+2}(=16)$ **or** $\frac{180-4}{11}(=16)$		4	M1	for using the angle sum on a line to form an equation in x **or** for a numerical method to find x using the angle sum on the line
	$2\times"16"-23\ (=9)$ **or** $9\times"16"+27\ (=171)$			M1	for using the value of x to find the exterior or interior angle
	eg $\frac{360}{n}="9"$ oe **or** $(n=)\frac{360}{"9"}$ $\frac{180(n-2)}{n}="171"$ oe **or** $(n=)\frac{360}{180-"171"}$ oe			M1	for using the formula for an exterior or an interior angle to form an equation in n **or** for a complete method to find n
	Correct answer scores full marks (unless from obvious incorrect working)	40		A1	
					Total 4 marks

Question 22 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 3*

3	$ADC = 180 - 58\ (= 122)$ **or** $EDF = 122$ **or** $CDE = 58$ **or** $ADF = 58$			M1	may be seen marked on the diagram
	e.g. $DEF = 58 \div 2$ **or** $DEF = (180 - 122) \div 2$			M1	complete method to find angle DEF
		29		A1	
			5	B2	dep on M2 for fully correct reasons for their method (B1 dep on M1 for one correct reason stated and used) e.g. Allied angles, co-interior angles, Alternate angles, Corresponding angles, Vertically opposite angles are equal (or Vertically opposite angles are equal), Angles on a straight line add up to 180°(or angles on a straight line add to 180°), Sum of two angles in a triangle are equal to opposite exterior angle, Angles in a triangle add up to 180°(or Angles in a triangle add up to 180°), Base angles in an isosceles triangle Angles in a quadrilateral add up to 360. (accept "4-sided shape"or parallelogram) Opposite angles of a parallelogram are equal
					Total 5 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 14

14	(a)	$(0.5\times)\ 9.3\times 14.7\times \sin 106$ **or** $(9.3\times\cos 16)\times 14.7$ **or** $(9.3\times\sin 74)\times 14.7$		2	M1 for applying the area of a triangle formula using correct values (to find half of the area of the parallelogram) **or** for a correct method to find the area of the parallelogram
			131		A1 awrt 131
	(b)	$(GE^2=)\ 9.3^2+14.7^2-2\times 9.3\times 14.7\times\cos 106$		3	M1 for the correct use of the cosine rule
		377(.9….) **or** 378 **or** 86.49 + 216.09 + 75.3… **or** 302.58 + 75.3….			M1 (dep on M1)for the correct order of operations
			19.4		A1 for 19.4 – 19.5
					Total 5 marks

Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2020 · Question 11

11	$(AX=)\ (17.6-8.4)\div 2\ (=4.6)$		6	M1 where X is the foot of the perpendicular from B to AD
	$0.5\times(8.4+17.6)\times h=179.4$ **or** $0.5\times$ '4.6' $\times h+0.5\times$ '4.6' $\times h+8.4\times h=179.4$ **or** $13\times h=179.4$			M1
	$(h=)\ 179.4\div$ '13' (=13.8) **or** $(h=)\ 358.8\div$ '26' (=13.8) oe			M1
	$\tan ABX=\dfrac{\text{'4.6'}}{\text{'13.8'}}$ **or** $\tan BAX=\dfrac{\text{'13.8'}}{\text{'4.6'}}$			M1 ft their h dep on second M1 $(AB=)\sqrt{\text{'4.6'}^2+\text{'13.8'}^2}=\sqrt{211.6}$ $=(14.546\ldots)$ **and** one from $\sin ABX=\dfrac{\text{'4.6'}}{\text{'}\sqrt{211.6}\text{'}}$ or $\sin BAX=\dfrac{\text{'13.8'}}{\text{'}\sqrt{211.6}\text{'}}$ or $\cos ABX=\dfrac{\text{'13.8'}}{\text{'}\sqrt{211.6}\text{'}}$ or $\cos BAX=\dfrac{\text{'4.6'}}{\text{'}\sqrt{211.6}\text{'}}$ or $\sin ABX=\dfrac{\text{'4.6'}\times\sin 90}{\text{'}\sqrt{211.6}\text{'}}$ or $\cos ABX=\dfrac{\text{'}\sqrt{211.6}\text{'}+\text{'13.8'}^2-\text{'4.6'}^2}{2\times\text{'}\sqrt{211.6}\text{'}\times\text{'13.8'}}$
	$(ABX=)\ \tan^{-1}\left(\dfrac{\text{'4.6'}}{\text{'13.8'}}\right)(=18.4)$ **or** $(BAX=)\ \tan^{-1}\left(\dfrac{\text{'13.8'}}{\text{'4.6'}}\right)(=71.6)$			M1
		108.4		A1 awrt 108.4
				Total 6 marks

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